



**G20 RESEARCH AND INNOVATION
WORKING GROUP**

**MPUMALANGA REPORT: GLOBAL COLLABORATION
SUPPORTING MUSEUMS AND NATURAL HISTORY
COLLECTIONS**

PRIORITY 2: BIODIVERSITY INFORMATION FOR SUSTAINABLE DEVELOPMENT

**DELIVERABLE 2.2 – NATURAL HISTORY COLLECTIONS COLLABORATION SEMINAR
AND WORKSHOP**

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ACRONYMS

Acronym	Full Name / Description
AI	Artificial Intelligence
AU	African Union
ARC	Agricultural Research Council (South Africa)
BGE	Biodiversity Genomics Europe
BID	Biodiversity Information for Development
CARE	Collective Benefit, Authority to Control, Responsibility, and Ethics (principles for Indigenous data governance)
CBD	Convention on Biological Diversity
CETAF	Consortium of European Taxonomic Facilities
CHF	Swiss Franc
COP	Conference of the Parties
CITES	Convention on International Trade in Endangered Species
CSRD	Corporate Sustainability Reporting Directive (European Union)
DAGI	Data Aggregator for Natural History Collections in Switzerland
DES	Digital Extended Specimen
DISSCo	Distributed System of Scientific Collections (Europe)
DSTI	Department of Science, Technology and Innovation (South Africa)
eDNA	Environmental DNA
EAf	Ecosystem Approach to Fisheries
EU	European Union
FAIR	Findable, Accessible, Interoperable, Reusable
FAO	Food and Agriculture Organization
FBIP	Foundational Biodiversity Information Programme
G20	Group of Twenty (International economic forum)
GBIF	Global Biodiversity Information Facility
GIDA	Global Indigenous Data Alliance
GRSciColl	Global Registry of Scientific Collections
iQoqwana	(A self-assessment and institutional strengthening tool developed by NSCF)
IUCN	International Union for Conservation of Nature
MCCs	Microbial Culture Collections
NHCs	Natural History Collections
NHM	Natural History Museum (Oslo)

NMK	National Museums of Kenya
NSCF	Natural Science Collections Facility (South Africa)
OECD	Organisation for Economic Co-operation and Development
Plazi	(Organisation supporting open-access to biodiversity taxonomic literature)
RIWG	Research and Innovation Working Group (G20)
SAIAB	South African Institute for Aquatic Biodiversity
SAMBA	Scaling Advanced Methods of Biodiversity Assessments
SANBI	South African National Biodiversity Institute
SANCF	South African National Collection of Fungi
SDGs	Sustainable Development Goals
SwissCollNet	Swiss Natural History Collections Network
TAP	Tropical African Plants
UN	United Nations
UNESCO	United Nations Educational, Scientific and Cultural Organization
UK	United Kingdom
WFCC	World Federation for Culture Collections
WIPO	World Intellectual Property Organization
Zenodo	(Open-access repository developed under the European OpenAIRE program and operated by CERN)

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SECTION 1

Introduction

The world faces an unprecedented array of global challenges, including climate change, biodiversity loss, and emerging diseases. Global biodiversity is in a state of unprecedented decline, posing profound and escalating risks to ecological stability, economic prosperity, and human well-being. The Living Planet Index reveals a staggering 73% average reduction in monitored wildlife populations since 1970, with freshwater systems experiencing the most severe losses. This degradation is driven by a complex interplay of human activities, primarily habitat loss, overexploitation, pollution, and climate change, which often act synergistically to amplify adverse effects. The prevailing economic paradigm, which systematically undervalues nature and fails to account for the depreciation of natural capital in conventional metrics like GDP, has inadvertently fuelled this crisis. This systemic mispricing has led to a significant "Impact Inequality," where humanity's demands far exceed the biosphere's regenerative capacity¹. Natural History Collections (NHCs) are both repositories of invaluable biological information and are also a **distributed scientific infrastructure** with the capacity to actively engage with and support solutions-driven science. These **research infrastructures** are uniquely placed to address urgent and emergent issues through research, education, data delivery and conservation. Nevertheless, the full potential of these collections is often untapped due to various constraints, such as limited access, lack of digitisation, and limits to data access and sharing, limited staff capacity, and limited resources for individual institutions. The [Global Collections paper](#) of 2023 outlined a path towards addressing these challenges.

Natural History Museums and Natural Science Collections are becoming recognised as essential research infrastructure, documenting patterns of biological diversity and change across time. Beyond their scientific importance, these collections also have significant socio-economic value. They provide critical baseline data for sectors such as agriculture, bioprospecting, biosecurity, energy security, public health, tracing the origins of emerging diseases, helping to identify invasive species, and guiding effective land use practices. By supporting industries through accurate species identification and ecological forecasting, natural history collections contribute to risk assessment and mitigation and cost savings. Moreover, the digitization of these resources enhances and accelerates global accessibility and fosters innovation in biotechnology and conservation-based economies, reinforcing their role as a foundation not only for scientific advancement but for economically informed decision-making.

Purpose

The Seminar and Workshop ***“Developing Global Collaborations to Support Museums and Natural History Collections”*** was crafted within the framework of the G20 Research and Innovation Working Group, and was held on the 28th-29th May 2025, at the University of Mpumalanga. This event brought together more than 60 global participants (Appendix 1) and was considered a timely and strategic opportunity to address one of the most foundational elements of biodiversity science: **the collections that underpin our understanding of the natural world.**

This G20 Seminar and workshop is about partnerships. It's about action. We recognise that some of the largest NHCs and associated Research Infrastructures reside in the Global North, with much of the world's high biodiversity regions occurring in the South. **Through this Seminar aimed at Global Collaboration**, we

¹ <https://www.livingplanetindex.org/>

want to ensure that natural history collections — many of which reside in institutions across the Global South — are empowered, accessible, digitised, and integrated into the global scientific infrastructure.

The aim of the Seminar and Workshop is to ultimately produce the “Mpumalanga Roadmap for Advancing Global Collaboration in Natural History Collections”, outlining **potential collaborative opportunities** which will be submitted to the RIWG processes and for inclusion in the Ministerial meeting in September 2025. This will form part of the Pretoria Package and will be referenced in the Ministerial Declaration.

Natural History Collections, the Organisation for Economic Co-operation and Development (OECD) and GBIF

In 1999, leaders of the global natural history collections (NHC) community recognised the urgent need for international collaboration to better utilise the vast and fragmented biodiversity data held in collections around the world. With the support of the OECD’s Mega Science Forum, these visionaries—including predecessors of many present-day stakeholders—developed a comprehensive proposal outlining the creation of a global infrastructure to mobilise and share biodiversity data.

This proposal was presented to national ministries and science authorities—many of whom are represented today through the G20 Research and Innovation Working Group (RIWG). Their support led to the founding of the Global Biodiversity Information Facility (GBIF) in 2001, marking a major milestone in coordinated global action for science, research, and innovation in biodiversity.

Nearly 25 years on, GBIF has evolved into a mature and inclusive infrastructure that now mediates a wide array of biodiversity data types. Much of this progress has been achieved through continued collaboration with, and leadership from, natural history collections. NHCs remain dynamic institutions at the forefront of biodiversity science—driving innovation through the adoption of new technologies, the expansion of data types, and the strengthening of collaborative research networks.

As we reflect on our progress in 2025, it is clear that while much has been accomplished, the community must continue to expand its ambition and global impact. NHCs, in partnership with GBIF and other linked infrastructures, are well-positioned to deepen collaboration and address pressing global challenges, particularly the escalating biodiversity crisis. This event offers a timely opportunity to reinvigorate global scientific collaboration and to reassert the value of coordinated biodiversity infrastructure. It may also be time to revisit and update the original vision from 1999. With the renewed and expanded support of organisations such as the OECD and UNESCO, along with newer strategic allies like the G20 RIWG and the Convention on Biological Diversity (CBD), the global NHC community has a critical role to play in developing a forward-looking plan that demonstrates how collective action can more effectively respond to today’s urgent biodiversity challenges.

The Value of Biodiversity Data and Biodiversity Informatics

At the heart of the work of the biodiversity science community lies biodiversity data — the records, observations, specimens, and associated knowledge that tell the story of life on Earth. The field of biodiversity informatics enables us to manage, mobilise, analyse, and share this data at scale. It transforms static collections into a dynamic, digital infrastructure for science, conservation, and innovation.

Through biodiversity informatics, we can:

- Track the spread of invasive species and pathogens.
- Model the impacts of climate change on ecosystems and species.

- Support sustainable land-use planning and natural resource management.
- Inform global agreements like the Post-2020 Global Biodiversity Framework.
- Accelerate the discovery, description, and cataloguing of global species diversity.
- Establish critical baselines and long-term monitoring programs for biodiversity and planetary change.
- Advance genetic and genomic research for biodiversity conservation.
- Support public health and disease surveillance.
- Enhance educational initiatives and public understanding of science.
- Facilitate international and interdisciplinary collaboration by promoting shared data infrastructure, standardised practices, and capacity building.

Importantly, biodiversity data is not just about science — it is about **decision-making**. When made open, FAIR (Findable, Accessible, Interoperable, and Reusable) and inclusive, these datasets empower governments, researchers, Indigenous communities, and the public to take informed action.

That’s why investments in digitisation, data infrastructure, and data governance must be central to our future strategies. To fully realise this potential, we must invest not only in infrastructure but also in **capacity development** — training the next generation of biodiversity professionals, like data scientists, curators, taxonomists, and ecologists, is critical. Strengthening human capacity, especially in under-resourced regions, is essential for bridging data gaps, sampling bias, and ensuring long-term impact. Developing Centres to support capacity development is critical to enable the use and relevance of biodiversity data to respond to the biodiversity crisis.

South Africa’s Commitment & Role

South Africa holds the G20 presidency in 2025 and is proud to have been a host of this Seminar and Workshop, and to facilitate this engagement and dialogue. It is home to world-renowned biodiversity and also holds significant scientific infrastructure and collections through institutions like SANBI, the Iziko Museums, and the institutions that are part of the Natural Science Collections Facility (NSCF). We recognise the critical role that collections play in informing research, guiding conservation, supporting climate adaptation, and even enabling innovation in the bioeconomy.

The Need for Global Collaboration

As we face planetary-scale challenges — biodiversity loss, climate change, and growing inequality — collaboration is no longer optional. It is essential. We must work to **rebalance historical asymmetries in capacity, resources, knowledge, and skills** to ensure scientific relevance and visibility. This means elevating the voices, knowledge, and priorities of institutions in the Global South. It also means recognising and integrating Indigenous and local knowledge systems — not as supplementary, but as equally valid and vital.

Workshop

The aim of the workshop on the 29th May was to advance the outcomes of the Seminar. Many countries across the world have sent their NHC experts to support the seminar and workshop, including those from Asia, Europe, and Africa (see Appendix 1). This shows a commitment towards working and growing together in this field of work.

Objectives and Framing the Dialogue for the Seminar

The aim of the sessions of the Seminar was to:

- Reflect on the current state of natural history collections globally.

- Explore the role of technology and digitisation in unlocking new frontiers of biodiversity research.
- Examine ethical considerations, especially regarding data sovereignty and Indigenous knowledge.
- And most importantly, identify practical steps to strengthen global collaborations that are equitable, inclusive, and forward-looking.

SECTION 2.

G20 Research and Innovation Working Group (RIWG) Seminar Developing Global Collaborations to Support Museums and Natural History Collections 28th May 2025

Welcome and Opening

The Seminar was opened by Mr. Shonisani Munzhedzi, CEO of the South African National Biodiversity Institute (SANBI) and Head of Delegation to the GBIF Governing Board, and Dr. Joe Miller, who provided welcome and opening remarks. The programme for the seminar and workshop was supported by professional organisational transformation facilitators Ms. Ilze Olckers and Ms. Desiree Paulsen (Tamarind Tree Associates).

The half-day seminar included two sessions, with short keynote speeches and panel sessions (Appendix 2). The speakers and panellists were from South Africa, G20 countries, and other African nations.

Session 1: Global Collections: The Imperative for Action Now

Purpose: This session set the scene for the current state of play in natural history collections (NHCs). It looked at existing initiatives, activities, priorities, and challenges, including the value of collections to science, society, and the bioeconomy. The session identified some of the asymmetries outlined in the G20 RIWG deliverables (and the background document) and identified potential opportunities to forge new partnerships.

Session 2: Global Collections: Securing Our Future, Starting Today

Purpose: Natural history collections, as repositories of invaluable biological information and as distributed scientific infrastructure, with the capacity to actively engage with and support solutions-driven science and actionable knowledge, are uniquely placed to address urgent and emergent issues through research, education, and conservation. This session aimed to identify collaborative actions to help support global collaboration in the Natural History Collections arena and included topics like Indigenous Knowledge Systems and their applications in NHCs future thinking.

For the seminar, speakers highlighted several key initiatives, and some common themes emerged from the presentations and panel discussions. Some of the initiatives and themes are detailed in the section below:

1. Global Collections Collaboration for NHCs and the G20 Opportunity
2. Lessons from South Africa's Natural Science Collections Facility
3. Advancing Biodiversity Informatics: Global Collaboration for Sustainable Impact
4. Perspectives from Africa (opportunities and challenges)
5. Forward-looking approaches for NHCs into the 21st Century: Scalable, Open, and AI-ready Workflows for Biodiversity Discovery

Overall, several opportunities and challenges were also identified. A full list of recommendations based on the outcomes of the Seminar and Workshop can be found in Table 1.

Initiatives and Common Themes Emerging from the Seminar Presentations

A Global Collections Collaboration for NHCs and the G20 Opportunity

Dr. Tim Littlewood

Natural History Museum, London UK

Natural History Collections (NHCs), comprising over 3 billion specimens across more than 3,500 institutions globally, represent a critical but underutilised infrastructure for addressing some of the world's most urgent challenges—including climate change, biodiversity loss, and threats to planetary and human health.

These collections offer unparalleled long-term data essential for modelling environmental change, tracking disease emergence, and guiding conservation and sustainability efforts. The G20 RIWG has a unique opportunity to **catalyse international collaboration** by promoting **open access to NHC data**, supporting digital infrastructure like the **GRSciColl** and **innovation (e.g. genomics, AI, remote sensing)**, and embedding collections-based evidence into science policy.

Realising this potential will require addressing **challenges** such as data fragmentation, limited interoperability, and global capacity disparities. As a strategic global forum, the G20 can elevate NHCs within national and international research systems, ensuring that the evidence they hold informs equitable, data-driven responses to 21st-century planetary challenges.

As institutions, we must unlock, share, and connect the information and knowledge embedded within **global collections to understand the value of NHCs and the potential impact** in addressing both local and global challenges. The *Global Collections Approach* paper assessed 73 of the world's largest NHCs and identified NHCs at large as a *distributed scientific infrastructure* of critical importance. There was a need expressed to extend this approach, to include more countries from the South. The study also found that many institutions are making their specimen data available to the Global Biodiversity Information Facility (GBIF) and other infrastructures (like the Distributed System of Scientific Collections (DiSSCo) and the Atlas of Living Australia (ALA)), enhancing their relevance and visibility, particularly to those countries and regions from which the collections originated. GBIF is increasingly recognised as a unifying platform—making data discoverable, helping identify where information is held, and enabling meaningful coordination across institutions. When countries and institutions coordinate efforts and share data, the benefits extend beyond scientific discovery—they also generate significant economic value. This underscores the urgent need for cross-country dialogue and collaboration.

It is time now to **partner strategically and equitably**, to realise access and benefits from the **global collection** whilst identifying **how and what we collect today for tomorrow**.

To harness the power of NHCs, successful partnerships will need to be put in place. These partnerships will need to 1. Establish clear objectives and shared goals 2. Foster open communication and trust 3. Utilise existing networks and platforms 4. Develop **roadmaps and action plans** 5. Focus on capacity building and 6. Promote inclusivity and equity.

Lessons from South Africa’s Natural Science Collections Facility

Dr. Anthony Magee,

The South African National Biodiversity Institute

South Africa’s Natural Science Collections Facility (NSCF) has developed a coordinated national network of 19 institutions to address fragmentation, orphaned collections, resource constraints, legacy exclusion, and the need for public value in research infrastructure. Established in 2017 under the South African Research Infrastructure Roadmap, the NSCF promotes collaborative, cost-effective management of the country’s diverse natural science collections and has a strong focus on transformation.

Key initiatives include the development of the NSCF Manual, implementation of standard imaging protocols, including 3D protocols, virtual access platforms, and real-time self-assessment tools. The iQoqwana assessment framework is a participatory tool for strengthening institutional management and curation practices and tracking improvements over time.

Since its inception, the NSCF has improved documentation, migrated data to Specify software, increased the digitising of collections, and increased readiness for GBIF publication and use. The size, scope, and impact of the collections of the NSCF were elaborated. **Looking ahead, global collaboration, digital innovation, and equitable access remain central to maximising the scientific and societal value of South Africa’s natural science collections.**

Future opportunities include **global collaboration** on type specimen imaging, leveraging **emerging technologies for mass digitisation**, and enhancing **data analysis capacity** supporting research and policy priorities.

Natural History Collections for the 21st Century: Scalable, Open, and AI-ready Workflows for Biodiversity Discovery

Dr. Christoph Haeuser

Global Biodiversity Information Facility (GBIF)

Natural history collections, while foundational to biodiversity science, remain biased and incomplete—especially for small, understudied invertebrates. This presentation explored how scalable, open, and AI-ready workflows can modernize biodiversity discovery. It highlighted the severe data imbalance, with birds dominating global records while insects remain vastly underrepresented.

To address this, the **“Biodiversity Knowledge Factory”** was introduced, a system integrating **robotic digitisation (DiversityScanner), nanopore DNA sequencing, AI-based species identification, and automated taxonomy tools like Description**. These innovations enable rapid, cost-effective processing of large biodiversity samples, particularly from neglected groups.

The presentation also emphasized the importance of open data sharing through platforms like GBIF, Plazi (<https://plazi.org/>), and Zenodo, and showcased novel storage solutions for managing millions of specimens. By prioritizing “dark taxonomy”—the study of small, ecologically vital species—they argued for a shift in focus that could transform ecosystem monitoring and conservation science.

Advancing Biodiversity Informatics: Global Collaboration for Sustainable Impact

Dr. Fatima Parker-Allie

South African National Biodiversity Institute

Natural history collections (NHCs) represent critical scientific infrastructure for addressing biodiversity loss, climate change, and sustainable development. These collections document historical and present-day patterns of biological diversity, offering unparalleled insight into ecological and evolutionary processes. However, stark asymmetries persist between regions rich in biodiversity (primarily in the Global South) and those with the infrastructure, funding, and capacity to mobilise and utilize biodiversity data (largely in the Global North).

Efforts led by global platforms like the **Global Biodiversity Information Facility (GBIF)**, alongside national nodes, are addressing these imbalances by promoting data governance frameworks grounded in the **FAIR (Findable, Accessible, Interoperable, Reusable)** and **CARE (Collective Benefit, Authority to Control, Responsibility, and Ethics)** principles. These initiatives are enhancing the discoverability, accessibility, and ethical use of biodiversity data, while fostering global collaboration. South Africa, as host of the G20 in 2025, is positioned to champion equitable approaches to biodiversity informatics and NHC governance.

A landmark study of 85 million botanical specimens reveals ongoing colonial legacies in specimen distribution, with significant mismatches between where biodiversity exists and where collections are held. This disconnect reinforces the urgency of equitable digitization efforts and capacity building.

South Africa's efforts, led by **SANBI-GBIF**, demonstrate strategic investment in digitization and data infrastructure, including a **mass digitization conveyor belt system**, to mobilise data at scale, in collaboration with international partners, and are strengthening South Africa's role in the global data landscape. National programmes like the **Foundational Biodiversity Information Programme (FBIP)** and the **Natural Science Collections Facility (NSCF)** support these endeavours, with key policy drivers such as the DSTI White Paper on Science, Technology and Innovation underpinning this work. These efforts support the drive towards a knowledge-based economy, linking biodiversity data with innovation, decision-making, and policy outcomes (e.g., marine protected areas).

Despite being one of the most biodiverse continents, **Africa contributes only 2.5% of GBIF's digitized biodiversity data**, revealing urgent capacity gaps in data mobilisation and use. Capacity-building through **training, curriculum development**, and a proposed **distributed Centre for Biodiversity Informatics** is seen as a vital solution. Such **Centres for Biodiversity Informatics** can catalyse postgraduate education, data-intensive research, and policy engagement.

Central to this vision, and aligned to the RIWG outcomes, is the development of a **roadmap for regional and global collaboration**, to reduce capacity and knowledge asymmetries, deepen participation from underrepresented regions, and align biodiversity data efforts with sustainable development goals. This roadmap can guide collective action, ensure balanced representation, and build a strong foundation for future innovation and impact.

Perspectives from Africa

Contributions from Dr. Wayne Florence from (Iziko Museums of South Africa), Dr. Alex Asase (University of Ghana), and Dr. Geoffrey Mwachala (National Museums of Kenya)

Challenges: Africa, despite being one of the most biodiverse regions in the world, continues to face significant challenges in building and sustaining natural history collections (NHCs). These challenges are rooted in colonial legacies and are compounded by limited government funding, inadequate infrastructure, and a shortage of curation expertise and scientific capacity. Diminishing international collaboration—often due to the gap between policy intent and operational realities—further constrains progress. Regionally, cooperation remains limited, shaped by historical fragmentation and uneven development within the region. In addition, gaps in knowledge sharing among African institutions hinder the consolidation of expertise and resources. While advances in information and communication technologies offer new opportunities for digitization and data sharing, the continent’s ability to fully leverage these innovations is still constrained by systemic underinvestment and fragmented efforts.

Most **type specimens** are held outside the continent, restricting local access. Addressing this imbalance requires investment in digitization, training, ethical collaboration, and infrastructure. However, African museums face persistent funding, political, and institutional challenges that hinder their development and public engagement.

Further challenges were identified as follows:

Chronic Underfunding and Weak Infrastructure

Many institutions suffer from sustained underinvestment, resulting in outdated technologies, inadequate facilities, and limited operational capacity. The constant trade-off between basic maintenance and innovation hampers the ability to modernize and scale data systems or digitization efforts.

Skills Gaps and a Lack of Data Standards

There is a widespread shortage of expertise in biodiversity informatics, taxonomy, and collection management. Data remains siloed and inconsistently structured, with an absence of common standards. Additionally, limited training and development opportunities restrict institutional growth and resilience. Technologies like genomics, robotics, artificial intelligence, machine learning, and computational (modelling) skills are allowing other nations to unlock their collections more rapidly, thus emphasizing the need for basic and advanced skills development.

Policy Misalignment and Weak Governance

Disconnects between government departments, the absence of dedicated policies for digital heritage and biodiversity data, and unclear institutional mandates weaken strategic coordination. This fragmentation undermines long-term planning, resourcing, and integration of biodiversity data into national agendas.

Barriers to Integrating Indigenous and Local Knowledge

Historical biases in classification systems and **lingering colonial legacies** have marginalized Indigenous perspectives. The lack of appropriate ethical and legal frameworks, along with limited trust and inclusive governance structures, continues to obstruct the equitable inclusion of traditional knowledge systems in biodiversity science.

Mitigation Measures and Opportunities in Africa:

Initiatives like the [Tropical African Plants](#) (TAP) Project (lead by Prof. AT Peterson from the University of Kansas) and support from global platforms such as GBIF are helping to bridge data gaps and build capacity. Initiatives like the Biodiversity Information for Development (BID) have been instrumental on the African continent to mobilise data and grow capacity. Ethical, inclusive collaboration, enhanced regional cooperation, technological integration, and data sharing are key to unlocking the full potential of Africa's biodiversity resources. Recognition of indigenous knowledge has been identified as critical.

Flagship Museums on the continent include the National Museums of Kenya (NMK), with over **6.4 million specimens** and approximately 140 scientists. The NMK supports international obligations and is Kenya's scientific authority for the CBD and Convention on International Trade in Endangered Species (CITES), and is also the National Focal Point for the Ramsar Convention. It also hosts the Kenya node of GBIF and is a data publisher, making 27 datasets accessible in 2025.

Iziko Museums celebrates its 200th anniversary in 2025 and aims to evolve into a world-class African institution by **integrating science, culture, and Indigenous knowledge** to tackle contemporary challenges. It has a strategic focus on exhibitions, collections, science communication, capacity building, and humanity, with **> 10 million specimen lots**.

Further Opportunities were Identified:

Enhance Technical and Financial Support

Foster partnerships with global institutions to co-finance the digitization of collections, upgrade physical and digital infrastructure, and expand access through open data platforms. Long-term, sustainable investment is essential to modernise systems and unlock the value of Africa's biodiversity data.

Invest in Capacity Building and Training

Establish regional exchange programmes, mentorship schemes, and accessible online training focused on biodiversity informatics, data standards, digital curation, and collections management. Strengthening technical skills across institutions will be critical to closing knowledge and capacity gaps.

Align Policies and Governance Frameworks

Collaborate with international/regional partners to develop and share best practices and integrated policies that support digital heritage preservation and biodiversity data governance. Promoting alignment across government departments and institutions will help ensure coherent and effective national strategies.

In the context of South Africa's Natural History Collections, an opportunity exists to more strategically align NHCs to fit under the DSTI. This will support the recognition of NHCs as critical research infrastructure, which is fully supported under the policy framework of the DSTI. This governance arrangement will strengthen the ability to secure appropriate funding and garner recognition to effectively safeguard the collections and make them accessible for science and society.

Promote Ethical Engagement and Inclusive Knowledge Systems

Support the co-creation of data initiatives with Indigenous and local communities, ensuring that ethical frameworks are in place to guide respectful engagement, equitable benefit sharing, and the protection of sensitive knowledge. This includes addressing historical imbalances in how biodiversity knowledge is classified, used, and governed.

Feedback from Participants on Asymmetries in NHCs

Bridging Asymmetries: Building Equitable and Sustainable Partnerships for the Future of Natural History Collections

To harness the collective expertise of workshop participants, a questionnaire was circulated to gather feedback on key aspects related to asymmetries in natural history collections (NHCs). The aim was to better understand perspectives and experiences. The questions posed were:

- What do we know and are we learning about the sustainability of the Natural History Collections themselves?
- What might contribute to more compelling stories about the critical role and value of Natural History Collections during the planetary crisis we are facing?
- What does Solidarity mean to us in the context of our Global Collections?
- What is being revealed to us about the asymmetries, imbalances and ethics in Natural History Collections?
- What do we need to keep in mind when developing global collaborations to support Natural History Collections?



What do we know and are we learning about the sustainability of the Natural History Collections themselves?

- Governments also need to commit to strengthen Museums, look at current platforms and networks to facilitate collaboration and cooperation.
- The understanding of sustainability and dynamic nature of NHC is well known and documented. So that is not questionable, nor is resources required etc. what is an issue is inconsistent and misdirection of funding. For example smaller collections receiving more funding and resources than larger ones.
- We know that we have to protect and curate the collections we have, discover and make openly accessible dark information on specimens and taxonomy, make information openly accessible through global infrastructures such as GBIF. But we also have to collaborate in working out strategies for new collections, specimen data and taxonomic knowledge for the future. This should not only be tagged by each country itself, but should be done in common and be globally coordinated.
- That it requires broader cooperation and collaboration between the global north and south.
- That we cannot sole rely on government/public sector to 'save' NHC. We need buy in from the private sector and provide them with tangible benefits of investing to NHC.



Bridging Asymmetries: Building Equitable and Sustainable Partnerships for the Future of Natural History Collections

- The mobilisation of data from NHCs is critical to enable use and impact. Open access is critical, however this also needs to ensure attribution to the relevant data publishers. Also the ability to link data types from different systems in an integrated manner is now also happening, 2. the development of case studies.
- Research stemming from these collections and communication of that, not only to an elite scientific few but to general public is essential. This can be done in an inclusive manner and also real life wow shock findings used to showcase the importance such as the discovery of a new species in the NHC which during our lifetime has since gone extinct. This is real time so to speak and will get attention as to the importance as well as need for such facilities.
- Good narratives, show, what you have and underline the functions and importance of selected specimens with easily understandable stories. Open doors to collections for policymakers and the public. Appeal to the protection of national cultural goods. Showcase good practices and investments of other countries.
- The development of an economic model that illustrates very clearly the value of NHC to science and society.



What might contribute to more compelling stories about the critical role and value of Natural History Collections during the planetary crisis we are facing?



What does Solidarity mean to us in the context of our Global Collections?

- Being able to work together in a manner that is respectful and inclusive. The breaking down of silos is important. We need to talk. Interesting is the openness and willingness of invited partners to collaborate and work together.
- Sharing resources, expertise and efforts to answer shared questions, goals and find solutions is a must. This, collaboration is essential.
- Open access to specimen data help reconstruct biodiversity and geodiversity changes over time, the more data, the more reliable data models can be made. Transparency and open access to methodological knowledge and data. If we share our knowledge, we do not have to invent things multiple times.
- Collaborations and knowledge exchange help to apply standardized methods for collection management, improvements of collections quality and data aggregation for common use.
- Meta-data generation of collections and mass digitization, apply FAIR data principles.
- Sharing infrastructure and mobilizing biodiversity data and expertise to solve urgent global challenges like climate change, pandemic preparedness, and biodiversity loss that cannot be tackled at a regional scale alone.
- To provide actionable targets that are realistic, given the differences in capacity and infrastructure availability between the global north and south.



Bridging Asymmetries: Building Equitable and Sustainable Partnerships for the Future of Natural History Collections

- The reality of the imbalances of where the collections reside and where biodiversity has been collected, and the very rich biodiversity that still lives in the global south. Africa wide consortium, priorities at continental level, biodiversity groups what are their strengths and what do we need to prioritize. Edwin mentions also a regional approach - Asia, Etc. Capacity in country to answer national and international priorities.
- Previous practices are undoubtedly flawed. However, focusing on this rather than the current operations could misdirect attention and efforts to what is actually needed. But even with national systems in place, there can still be bias in the allocation of funding and resources.
- Seek collaborations with the South, when research projects have focus on species from their countries.
- Open access to metadata of collections help the South to locate indigenous specimens and establish collaborations and eventual restitutions of indigenous collections".
- That these are aspects that museums are confronted with globally, and that there is a sense of responsibility and urgency to implement corrective measures and bridge socio-economic and political divides to safeguard and provide access to collections.
- That historical imbalances are being addressed and a lot of progress has been made as a result of collaborations between the north and south.



What is being revealed to us about the asymmetries, imbalances and ethics in Natural History Collections?



What do we need to keep in mind when developing global collaborations to support Natural History Collections?

- Human relationships, not to repeat the behaviour of the past. Ownership, agency and humility are important as we forge ahead.
- Collaboration can be complex, and collaboration usually has a hidden agenda. For example, many times collaborations are formed because someone wants access to something of their interest. They then form collaboration but it is purely for a specific individual purpose. If this is a research example then the other individual will receive co-authorship, for example. But do they actually benefit anything, and was the collaboration short-term to accomplish the goal of the initiator? Will the initiator continue past this after they get what they want. Institution level collaboration can be seen in a similar light and for a specific reason and timeframe.
- Ethics, respect different cultures, give the chance to everybody to participate. Open access to information for everybody, transparency.
- That the available infrastructure, expertise and capacity differences between the global north and south - skills transfer and capacity development should be prioritised.
- A Global collections initiative requires investment to bridge capacity and infrastructure gaps between museums in the Global North and South.



Research and Innovation Working Group Workshop

Developing Global Collaborations to Support Museums and Natural History Collections

29 May 2025, University of Mpumalanga, Mbombela, Mpumalanga Province

The Workshop focused on deepening the conversations from the Seminar, adopting a more purpose-driven, participatory approach, designed to identify short, medium, and long-term collaborative opportunities to inform the development of a global Roadmap for strengthening museums and natural history collections.

Opened by Dr Theressa Frantz, SANBI's Head of Branch, the workshop used a mix of keynote discussion starters, interactive formats, and collaborative reflection. Participants explored themes such as: 1. new roles for old institutions, 2. Developing networks to support biodiversity data mobilisation, and 3. Building inclusivity and transdisciplinary capacity, and to further address aspects of inclusivity and colonial legacies in biodiversity science, 4. Exploring Mechanisms Among Global Natural History Collections for Inclusive Community Consultation and Multidisciplinary Data (Including indigenous knowledge) Integration.

While the initial sessions of the day aimed to showcase key activities and initiatives within the natural history collections (NHCs) and biodiversity informatics landscape, the second half focused on distilling these highlights into actionable recommendations and clear outcomes (Table 1) to inform the development of the Roadmap. A session focused on *Integrating Infrastructure and E-Infrastructure to Advance Natural History Collections into the Digital Age* was also included during the latter part of the day. To further support this process, a questionnaire was provided to participants to help identify and capture potential collaborative opportunities. This information has been captured and integrated into the recommendations and outcomes, and also the Roadmap.

Workshop Feedback Emanating from Presentations and Facilitated Sessions

Sensing 1: New Roles for Old Institutions: Transforming the Museums and Natural History Collections Landscape

Keynote: Bending the curve through collaboration with policy and industry

Mr. Edwin Van Huis

Naturalis Biodiversity Center

The “shifting baselines effect” means that every generation only experiences biodiversity loss relative to its own experiences. Based on the current environmental crisis and the unprecedented rate at which we are losing species, our efforts remain insufficient. We are not making an impact in successfully fighting the biodiversity crisis because we have not succeeded in making it a priority for people. *We may have reached their heads, but we have not reached their hearts.* The power of the economy is a stronger driver of behavioural change. ***It's governments and the industrialized world that can effectively fight a biodiversity crisis.***

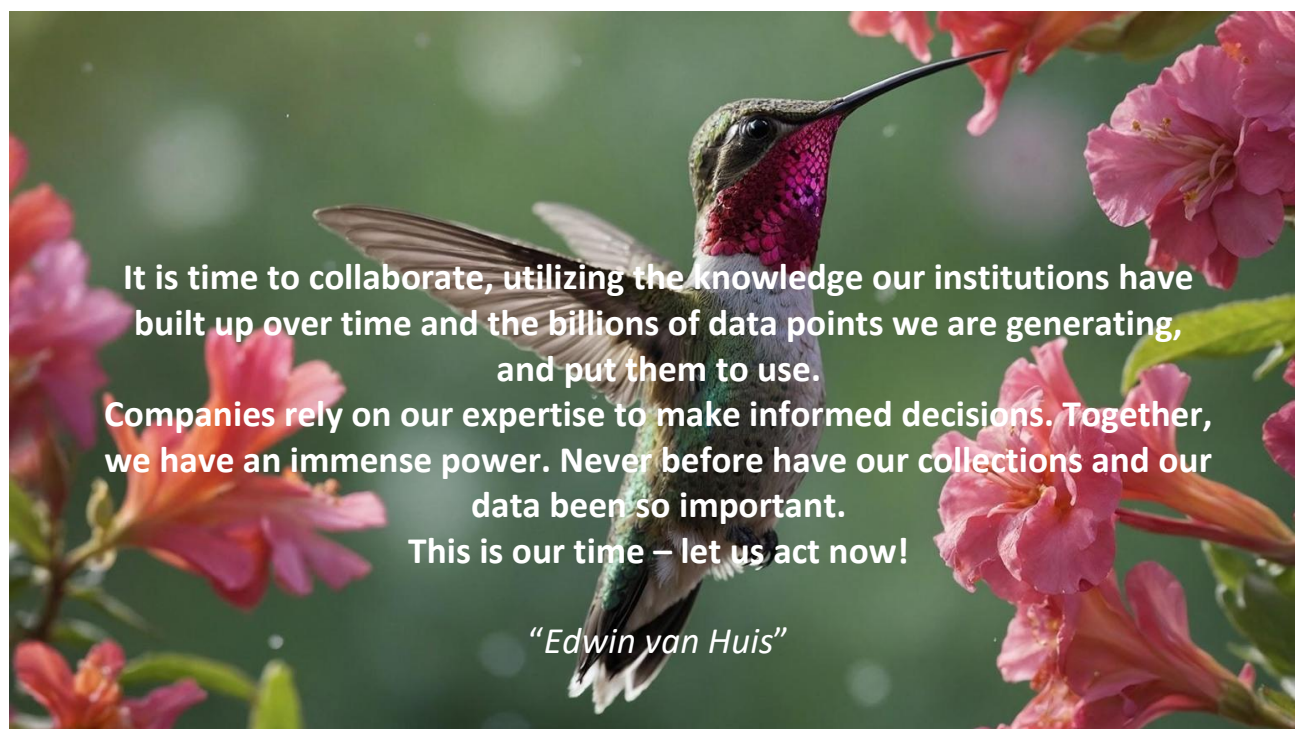


There are alignments between the climate crisis and the biodiversity crisis. **In a similar manner to the evolution of the climate crisis and its integration into world economics, the biodiversity crisis is becoming part of the economy.** A first step was taken in the financial sector, involving banks, insurance companies, and investment companies, who saw a world in which their investments were at risk. Climate risks and biodiversity risks were never part of the equation, and now they are. The European Central Bank is calculating these risks for European banks, and examples can be seen with the decline in **ecosystem services** directly impacting the balance sheets of banks, reaching up to a high dependency on ecosystem services in various EU countries, of up to 90%, with bank losses projected to rise to around 40% annually, under a business as usual strategy. So now banks are finding solutions to benefit their clients, and moving towards a nature-inclusive economy. To assess biodiversity risk and loss, the European Commission has issued a [Corporate Sustainability Reporting Directive \(CSRD\)](#), which requires companies to report, amongst other areas, on sustainability and climate. **Biologists have a unique opportunity to help demystify ecological complexity for the banking sector, supporting companies in understanding their obligations under the CSRD reporting requirements, guiding strategic investment decisions, and enabling investors to make informed choices that align with both financial goals and environmental responsibility.**

Naturalis Biodiversity Institute (in collaboration with others, e.g. CETAF, DiSSCo, Biodiversity-Meets-Data and Biodiversity Genomics Europe) has developed a model for biodiversity assessment that's relatively easy to use, and has offered to make it available to all for free. The model explicitly aims to work with open-source technology, shared data, and infrastructure throughout Europe, setting one standard for all users. The

outstanding achievement in the last decade in Europe is that NHCs institutions, research institutions, and universities have invested heavily in new technologies and biodiversity data. With new technologies and tools such as environmental DNA, AI, speech recognition, data science, scientists and community scientists collect millions of records, which are digitized into databases, and are contributing to GBIF. These layers (GBIF occurrences, IUCN red lists, remote-sensed data, eDNA etc.) are used in the biodiversity assessment tool in an integrated manner, **and provide companies with easy-to-use assessments and indicators to see how they can mitigate biodiversity loss in their operations and investments, and see which investments will benefit biodiversity or harm it.** The value-add in this example was that the tool/service is designed to ensure that **every company, regardless of its size or resources, has access to high-quality, transparent biodiversity data,** thus small companies can **assess and invest in their ecological impact** with the same clarity as the large corporations.

Biodiversity institutes should step out of their comfort zone and collaborate with the financial world and industry, as biodiversity is part of the business equation, and we need to help them navigate the complexities of this new approach.



New Roles for Old Institutions: Perspectives from the United Arab Emirates, the Kingdom of Saudi Arabia and Singapore

Contributions from Ms. Noora Albalooshi (Natural History Museum Abu Dhabi), Ms. Norah Alkhamis (Saudi Heritage Commission, UNESCO), Ms. Jia Jun Chua (Ministry of Trade and Industry-Singapore)

Around the world, museums and natural history collections (NHCs) are transforming from passive repositories into dynamic institutions at the forefront of biodiversity science, public education, and policy influence. As countries increasingly recognize the value of collections-based research and data-driven decision-making, they are modernizing these institutions to meet contemporary challenges—from sustainability and conservation to national development and global partnerships, to maximise their impact.

In the **United Arab Emirates**, the newly established **Natural History Museum Abu Dhabi** is pioneering a regional model for excellence in natural history. With world-class facilities and a strong emphasis on collections-based research, digitization, and access. The museum positions collections at the centre of everything it does. It aims to build sustainable, high-impact partnerships, and ***is actively seeking broad collaboration to advance biodiversity understanding and stewardship.***

In the **Kingdom of Saudi Arabia**, a national effort is underway to integrate culture, heritage, and science. Since the creation of the Ministry of Culture in 2019, multiple commissions have been tasked with revitalizing domains such as museums and heritage. A key initiative is a national digital archive, which aggregates collections from across institutions into a single, accessible platform for researchers. New museums, geoparks, and royal reserves are also being developed to bring stored collections into public view and align them with landscape-level conservation and education goals. International partnerships, like the archaeological **Green Arabia Project—involving 27 local and global partners**—are central to this mission, with the Saudi Heritage Commission placing mutually beneficial collaboration at the heart of its vision for impact.

Meanwhile, Singapore exemplifies how small nations can play a significant role through strategic, pragmatic investments in biodiversity infrastructure. Despite its size, Singapore **houses two major collections**: the **Lee Kong Chian Natural History Museum (LKCNHM)**, with over a million specimens, and the **Singapore Herbarium**, which holds approximately 800,000 specimens. These collections not only collect and digitize new material, but also reach beyond borders. Through initiatives like the Signify Project, LKCNHM collaborates with overseas museums to digitize specimens collected by explorers and naturalists over centuries. At SING, a major digitisation effort will complete a public online inventory of all herbarium specimens by 2027. These efforts help bridge data gaps and contextualize biodiversity trends across Southeast Asia, supporting both scientific inquiry and informed conservation policy. The SING also conducts regional capacity-building activities through staff attachment and workshops.

Together, these examples demonstrate how **"old institutions" are taking on new, globally relevant roles**. Whether through cutting-edge digital platforms, robust international partnerships, or local-to-global knowledge integration, countries are unlocking the full potential of their museums and collections. These efforts signal a collective shift toward natural history institutions that are more inclusive, accessible, and essential to understanding and safeguarding our planet's biological and cultural heritage.

Some Useful Comments and Discussions Emanating From The Session:

- **GBIF:** There is a Task Force on Nature-Related Financial Disclosures, working towards a Roadmap for Creating Market Access to Decision-Useful Nature-related Data. We, as the biodiversity sector, need to be at the table and to broaden that table once we understand what the private sector is doing. Especially, as there is a need from the markets to understand biodiversity and to sustain themselves. Understanding how the data can be used is going to be very powerful and relevant.
- **NSCF:** There is a disparity between the utilisation of the Northern Museums compared to Africa, and a suggestion to understand lessons learnt, and how to adequately foster public outreach and education in museums in Africa. This includes the use of technology and greater awareness raising, and including learner education at earlier stages.
- **DiSSCo:** We need to mobilise more parts of society. Based on our discussions, we note that the private sector has been a neglected stakeholder, although a very important one, as they are probably one of the most productive parts of society. There are now stronger compliance mechanisms, which have been put in place by the EU, that can be leveraged by the biodiversity sector.
- **International Centre for Innovation in Science, Technology and Education:** Even as we reflect on the inclusion of the private sector, there is also a deeper conversation about shifting the mindset of individuals to contribute towards biodiversity and planetary outcomes, which also starts at the grassroots level. Including indigenous communities, citizen scientists, and the public is also crucial as we venture into new spaces.

Sensing 2: Developing Initiatives and Building Collaborations at National, Regional, and Global Levels with a Focus on Data Mobilisation

Introduction

Natural history collections (NHCs) are vital for understanding biodiversity, environmental change, and ecological resilience. Yet despite their scientific and cultural importance, large parts of these collections remain inaccessible, underutilised, or disconnected from global data infrastructures. As biodiversity loss accelerates, the mobilisation and integration of data from NHCs have become an urgent priority. This report explores two complementary models of collaboration and innovation: the bottom-up, nationally coordinated initiative of SwissCollNet in Switzerland, and the distributed infrastructure approach of DiSSCo in Europe. Together, they illustrate how coordinated national, regional, and global strategies can unlock the full potential of natural history data.

SwissCollNet: A Bottom-Up National Initiative

Dr. Pia Stieger,
Swiss Academy of Sciences

SwissCollNet represents a successful example of bottom-up mobilisation of biodiversity data within a single country. Initiated by the Swiss Academy of Sciences, SwissCollNet emerged in response to longstanding calls from the collection's community for more resources, collaboration, and infrastructure. The programme was launched with CHF 12.3 million in federal funds, matched by institutional contributions, and aimed to strengthen natural history collections through digitisation, collaboration, and training.

Through competitive project calls, SwissCollNet funded 68 digitisation and curation projects that collectively involves 53 institutional units and more than 200 trainees. As a result, over 1 million specimens were digitised or reconditioned, and the quality of collection documentation improved significantly. The initiative also standardised metadata practices, resulting in 452 detailed collection profiles—representing over 61 million specimens—now accessible through the Global Register of Scientific Collections.

Crucially, SwissCollNet developed its own open-source data aggregator (DAGI), which maps local data to Darwin Core standards and publishes it to GBIF. The platform also supports hosted portals for both specimen and observational data. This has allowed Switzerland to unify its biodiversity data infrastructure, increasing the visibility and utility of national collections within global frameworks. The initiative also openly shares tools, documentation, and software developed through the programme, enabling reuse by other countries and institutions.

Switzerland is also playing a significant role in liberating data from scientific publications and developing novel ways for publishing in a semantically enhanced way (Plazi, Zenodo). This initiative is also the foundations for the **Disentis Declaration 2024**, a decadal roadmap for liberating global biodiversity knowledge from scientific literature.

DiSSCo: A Distributed Regional Research Infrastructure

Dr. Dimitrios Koureas

Naturalis Biodiversity Center

At the regional level, DiSSCo (Distributed System of Scientific Collections) offers a transformative model for data mobilisation through a distributed yet coordinated infrastructure. **Spearheaded by the Consortium of European Taxonomic Facilities (CETAF)**, DiSSCo brings together over **300 institutions across more than 20 European countries to create a unified, high-performance digital ecosystem for natural science collections.**

DiSSCo's vision centres on the development of high-throughput, end-to-end knowledge production systems that combine human expertise with cutting-edge technologies such as AI, low-cost DNA sequencing, and automated digitisation. The infrastructure is structured around four integrated programmes: digitisation, remote and transnational access, data and applications, and capacity building—especially targeting smaller institutions.

Central to DiSSCo is the concept of the **Digital Extended Specimen (DES): an enriched, machine-readable digital object that links specimen records with images, genetic data, geolocation, literature, and more.** DES supports both human researchers and automated agents, enabling collaborative annotation, species recognition, and data extraction at scale. This model addresses the uneven capacity across institutions by enabling centralised services while preserving decentralised collections.

Importantly, DiSSCo's work is grounded in the FAIR (Findable, Accessible, Interoperable, Reusable) data principles and aims to make collections accessible not only across Europe but globally. It aligns with efforts by GBIF, International Barcode of Life (iBOL), and the Ocean Biodiversity Information System (OBIS), among others, creating synergies with global platforms and accelerating data mobilisation in support of the 2030 biodiversity targets.

SANCF and African MCCs Consortium: Leading Fungal Data Mobilisation in Africa

Dr. Adriaana Jacobs-Venter

Agricultural Research Council

In Africa, microbial biodiversity is receiving increased attention through the efforts of the **South African National Collection of Fungi (SANCF)** and the emerging **African Microbial Culture Collections (MCCs) Consortium.** As presented by Adriaana Jacobs from the Agricultural Research Council (ARC), SANCF holds over 62,000 preserved specimens and 37,000 living cultures, with approximately 40% originating from outside South Africa, making it a unique pan-African asset. The collections also include living collections, biobanks, and the ARC contributes to the Natural Science Collections Facility (NSCF).

Operating within a complex institutional structure, the SANCF contributes to national mandates across agriculture, science and innovation, and environmental affairs. Its publication efforts began in 2017 through support from the GBIF South Africa node and the Natural Science Collections Facility (NSCF). These efforts included the development of a checklist for South African macrofungi and creative collaborations, such as an art project transforming fungal specimens into public-facing artworks. The collection also integrates rich associated metadata, including genetic sequences, morphological data,

and environmental samples (eDNA), while ensuring long-term preservation through dried voucher specimens.

Recognising the potential for economic impact, the ARC has begun commercialising fungal strains through partnerships with small and medium enterprises. Some strains are now licensed for agricultural use, and the SANCf is in the process of becoming a World Intellectual Property Organization (WIPO)-accredited repository for commercial strains — potentially the second such facility in Africa.

At a continental level, the SANCf has played a leading role in initiating the African Microbial Culture Collections Consortium, bringing together curators across Africa to address common challenges, ensure the safeguarding of microbial diversity, and increase the visibility and accreditation of African collections globally. A joint publication and proposal to the World Federation for Culture Collections (WFCC) Working Group on Endangered Culture Collections is currently in development.

Converging Lessons: Building Shared Infrastructure and Trust

The three case studies—DiSSCo, SwissCollNet, and SANCf—offer several shared lessons on mobilising biodiversity data through coordinated national and regional infrastructure:

- **Collaboration is essential:** No single institution can digitise and mobilise biodiversity knowledge at the required scale. National and regional consortia amplify capacity and enable equitable participation.
- **Interoperability is key:** Aligning with international standards (e.g., Darwin Core, FAIR principles) and integrating with platforms like GBIF ensures global usability of local data.
- **Capacity must be supported:** Smaller institutions and under-resourced regions require investment in digitisation tools, training, and long-term institutional support.
- **Distributed models work:** Federated systems that empower institutions to contribute data locally while linking nationally or regionally are both resilient and scalable.
- **Open science tools drive reuse:** Publishing aggregators (like DAGI), handbooks, and codebases ensures that infrastructure development is shared and sustainable.
- **Transformative technologies accelerate data mobilisation:** Adoption of transformative tools and technologies such as AI, models for image recognition, bioacoustics monitoring, and environmental DNA (eDNA) metabarcoding analysis. These innovations are unlocking new pathways for data mobilisation and discovery.

Conclusion

The need to mobilise biodiversity data has never been more urgent, with global biodiversity targets fast approaching and much of the planet's species still undocumented. National, regional, and global initiatives—from Europe to Africa—are demonstrating that strategic investment, open collaboration, and interoperable infrastructure can unlock the potential of natural and microbial collections.

By coordinating efforts and building platforms that are open, inclusive, and sustainable, the global biodiversity science community can transform collections into living, digital knowledge systems that inform conservation, policy, education, and innovation for decades to come.

Sensing 3: Building Capacity in the Natural History Collections Community to

Support Biodiversity Research and Transdisciplinary Collaboration

Contributions from Prof. Hugo de Boer (University of Oslo) and Mr. Roger Bills (South African Institute of Aquatic Biodiversity)

Natural history collections (NHCs) play a critical role in supporting biodiversity research, policy development, and cross-sectoral collaboration. Yet, across both the Global North and South, there are shared challenges: limited taxonomic capacity, unequal access to data, slow digitization rates, and imbalanced infrastructures. Recent initiatives led by institutions such as the Natural History Museum (NHM) in Oslo, Norway, and the South African Institute for Aquatic Biodiversity (SAIAB) provide important models for addressing these issues through targeted capacity building and international cooperation.

At the NHM Oslo, one key focus has been using collections not only as preserved archives but as reference points that can inform present-day environmental monitoring. For example, the SAMBA project (Scaling Advanced Methods of Biodiversity Assessments), developed in collaboration with Brazil, integrates museum collections with environmental DNA (eDNA) efforts to monitor and build biodiversity baselines. The SAMBA project aims to 1. train a new generation of scientists in cutting-edge genetic technologies and methods **to help speed up the description and monitoring of biodiversity** in Norway and Brazil; 2. techniques that can identify species from the DNA left in the environment can be used to **track species across space and time**. 3. Knowing what species are where, and how they respond to these changes is fundamental for making sound decisions on how to protect biodiversity. This is especially important in the context of the global taxonomic impediment, where undescribed species far outnumber those currently known. **By using eDNA and reference sequences from collections, researchers can detect biodiversity changes over time**—even when formal taxonomic work is incomplete.

Similarly, NHM Oslo's collaboration with GBIF through the BioDATA Phase I and II projects focuses on training early-career researchers in biodiversity data mobilisation and use. These workshops, held in countries like Armenia, Tajikistan, and South Africa (with SANBI-GBIF as a Partner), have taught students how to upload data to GBIF and leverage it in scientific research. This approach not only supports individual capacity development but **also promotes equitable access to biodiversity knowledge in underrepresented regions**.

These exemplary examples from Norway, of global collaboration in NHCs supporting research, training, capacity-building, and the use of these collections and associated data in support of science-based decision-making, are supported indirectly by the Norwegian Ministry of Education and Research through the Research Council of Norway and the Norwegian Directorate for Higher Education and Skills. These include the Natural History Museum, University of Oslo, led projects SAMBA and BioDATA.

On the African continent, the South African Institute for Aquatic Biodiversity (SAIAB) faces similar challenges in managing the vast amounts of material collected through active field expeditions, particularly through its involvement in the Food and Agriculture Organization's (FAO's) Ecosystem Approach to Fisheries (EAF) Nansen project. Every two years, enormous quantities of marine specimens arrive from cruises around the Western Indian Ocean. However, due to limited taxonomic staff and declining expertise, much of this material remains under-processed. In response, SAIAB hosted a marine fish taxonomy workshop in 2023, **pairing ten senior taxonomists with ten carefully selected young**

scientists from Kenya, Tanzania, and Mozambique. The focus extended beyond species identification to include literature reviews, genetics, data analysis, and scientific writing, ensuring participants gained end-to-end research experience. At present, about 20 papers from the workshop are in development, underscoring the productivity of this **mentorship-based model**. **SAIAB plans to institutionalize this format as an annual event**, expanding to neighbouring countries and gradually rebuilding the taxonomic community in the region. A similar workshop in collaboration with Ocean Census (UK) supported marine invertebrate training, showing the value of building expertise across taxonomic groups.

Both institutions emphasize the importance of sustained and inclusive capacity building. Success hinges on structured **partnerships, long-term funding, and the ability to demonstrate the relevance of biodiversity data to industry and policymakers**. There is a growing need to **create career pathways that connect biodiversity science with the private sector**, especially in areas such as environmental impact monitoring and nature-positive development. Museum collections must be continually renewed and connected to contemporary research and technology to remain relevant. Importantly, efforts must go beyond simply preserving existing specimens and toward building systems for ongoing data generation, sharing, and skilled human resource development.

In conclusion, **building capacity** in the natural history collections community requires a **coordinated strategy focused on inclusivity, innovation, and practical application**. By integrating training, data mobilisation, and collaborative research, institutions like the NHM Oslo and SAIAB demonstrate that collections can serve as living resources not just archives and support both biodiversity science and broader societal goals.

Sensing 4: Exploring Mechanisms Among Global Natural History Collections for Inclusive Community Consultation and Multidisciplinary Data (Including Indigenous Knowledge) Integration

Community Consultation and Indigenous Knowledge in Paleo Sciences

Dr. Wendy Black

Iziko Museums of South Africa

Iziko Museums has increasingly recognized the critical role of community consultation and the integration of indigenous knowledge in advancing ethical, inclusive, and impactful research in the paleosciences. Its recent work — including engagement with Khoisan communities on matters such as repatriation — has highlighted both the scientific and cultural value of meaningful partnerships with source communities.

These consultations have provided profound insights into traditional practices, contextual knowledge, and heritage narratives that have helped shape our research questions and methodologies. In particular, the integration of indigenous perspectives has supported new directions in paleo proteomic research, enabling us to target residues more precisely and interpret findings more holistically. Many of these outcomes would have been impossible without the community's deep, place-based knowledge.

This work has also helped to unlock dormant research potential in legacy collections. However, it has become clear that genuine inclusion of indigenous knowledge requires an ethical and strategic approach — one that avoids extractive models and instead fosters collaboration, co-production of knowledge, and mutual respect.

Iziko Museums emphasized two key points in this regard:

1. Ethical Integration of Indigenous Knowledge Systems

The inclusion of indigenous knowledge must go beyond simply adding it as an additional layer in scientific datasets. Without a clear ethical framework, this practice risks reinforcing extractive, Eurocentric research traditions. Responsible integration requires relationship-building, community consent, co-authorship, and the recognition of knowledge sovereignty and data ownership. As stewards of cultural and scientific heritage, we also acknowledge our responsibility to support the continued documentation and preservation of these knowledge systems, many of which are under threat. The failure to incorporate all available data — including indigenous knowledge — constitutes incomplete science. Furthermore, embracing diverse epistemologies strengthens research and encourages innovation through new ways of thinking and engaging.

2. Community Consultation as an Ongoing Process

At Iziko Museums, we view community consultation not as a checkbox activity but as a long-term, iterative process. Effective consultation involves time, trust, humility, and mutual benefit. While challenging, this process is essential to the integrity and relevance of our work. Community participation enriches scientific inquiry, and for communities, the benefits often lie in recognition, visibility, and the preservation of oral histories and cultural traditions — not merely in financial terms.

In conclusion, Iziko Museums reaffirms its commitment to a model of research that is collaborative, inclusive, and ethically grounded. By embedding community consultation and indigenous knowledge

into the heart of our scientific practice, we not only produce better research but also contribute to the transformation and decolonization of the museum and science sectors in South Africa and beyond.

Indigenous Data Governance: GBIF's Role and Emerging Collaborations

Dr. Joe Miller

Global Biodiversity Information Facility

Building on broader efforts to engage indigenous communities in biodiversity science, GBIF is taking key steps toward supporting indigenous data governance — focusing not on interpreting indigenous knowledge directly, but on how communities wish their data to be managed, shared, and attributed.

In response to recommendations from the Global Indigenous Data Alliance (GIDA), **GBIF is establishing a Task Force on Indigenous Data Governance**, with community-led nominations and membership from South Africa, North America, Sweden, Brazil, Australia, and Aotearoa New Zealand. The Secretariat has intentionally taken a hands-off approach to ensure the process reflects community priorities rather than top-down selection. The task force's Terms of Reference are being finalized, with an update expected at the Global Nodes Meeting in October.

A parallel pilot project is underway to implement **indigenous metadata labels**—such as **Biocultural and Traditional Knowledge Labels**—developed by the Grounding Indigenous Rights initiative. These labels allow communities to indicate culturally significant information (e.g., provenance, protocol restrictions, seasonal use) without disclosing sensitive knowledge and can be attached to natural history specimens in museum collections.

GBIF's role is technical: integrating these labels into its infrastructure so they can be aggregated and displayed on GBIF.org and hosted portals. Initial testing is being done with datasets from a few museums, and a demonstration portal is planned for the Living Data Conference. This will showcase how such community-driven metadata can be embedded in biodiversity data systems.

Ultimately, this work is about more than data—it reflects a shift towards ethical, inclusive, and community-governed biodiversity infrastructures, where indigenous voices help shape how their knowledge is represented and respected globally.

Integrating Infrastructure and E-Infrastructure: Advancing Natural History Collections into the Digital Age

Dr. Vincent Smith

Natural History Museum, London UK

This presentation highlighted the critical role of integrated infrastructure and e-infrastructure in transforming natural history collections (NHCs) into globally accessible, data-rich resources that underpin both science and society. The talk was framed within the urgent need **to build a global, digitally integrated natural history collection** that supports **sustainable development** and biodiversity policy, particularly in the context of COP15 and the UN Sustainable Development Goals.

1. The Case for Change and Global Integration

The talk referenced the pivotal “global approach to NHCs” paper co-authored by many in the community, which articulated the vision of a unified digital natural history collection. This vision promotes interoperability and accessibility to enable science-informed decisions for societal challenges such as biodiversity loss, agriculture, and health.

2. Driving Forces: Data Explosion, AI, and Collaboration

Three transformative forces were identified:

- **Data Mobilisation:** With over 3.2 billion biodiversity records already accessible (e.g., via GBIF), new machine-based observations are projected to exponentially increase this number. This demands scalable infrastructure and robust data curation systems.
- **Artificial Intelligence (AI):** AI presents unprecedented opportunities for automation, species identification, environmental modelling, and quality control — but requires significant investment in training datasets, taxonomic expertise, and equitable access to technology.
- **Partnerships and Infrastructure:** Collaboration across institutions and nations is essential. National infrastructures play key roles as mobilisers, standard-bearers, and enablers of data sharing.

3. Lessons from National Infrastructure: The UK DiSCCo Model

The United Kingdom’s experience in establishing DiSCCo-UK, a national initiative aligned with the broader DiSCCo Europe effort. Key elements included:

- Building a robust business case for digitisation.
- Coordinating over 90 institutions through a hub-and-spoke model.
- Unlocking **£155 million in funding for a 10-year digitisation programme** starting in 2026. Developing infrastructure to integrate NHCs, geocollections, and other mixed heritage collections.
- The DiSCCo-UK Hub includes about 91 institutions digitizing collections, with infrastructure supporting the storage and accessibility of data, sharing data with infrastructures like GBIF.
- Many gaps remain — notably for geological collections and human remains — and encouraged roadmap development to address these.

4. AI and Emerging Technologies: Proof-of-Concept to Impact

The presentation also detailed how the Natural History Museum (London) uses AI through a proof-of-concept model:

- AI applications included label transcription and specimen identification, specimen identification for customs enforcement, AI chatbots for data mining specimens, and robotic automation for digitisation.

- Robotics, drone technology, remote sensing, and the Internet of Things (IoT) were highlighted as vital for scaling digitisation.
- The ambition is to build a biodiversity-specific large language model (akin to ChatGPT), which will converge infrastructures and data, integrating observations, collections, literature, genetic, acoustic, and environmental data from collections to support solution-driven research.

The presentation also addressed the huge imbalances and asymmetries between those who have access to these technologies, for developing beneficial outcomes, and those who don't, including developing nations. Bridging those imbalances and addressing those asymmetries are critical. The potential solution could be the development of infrastructure-related projects, providing the glue that connects people, computing power, and skills development. It connects the data in such a way that you can start to deal with imbalances.

5. SYNTHESYS as a Model for Global Roadmaps

The presentation was concluded by revisiting the SYNTHESYS programme, a long-standing EU-funded initiative that supported access to collections, networking, and collaborative research across four phases. With over €38 million invested and thousands of researcher days funded, SYNTHESYS is a successful template for what a G20 roadmap for natural history collections could resemble — blending capacity building, infrastructure, and collaborative research at scale.

Key Takeaways for the G20 Roadmap

- Infrastructure — both physical and digital — is foundational for data mobilisation at national and global levels.
- There is a need to strengthen capacity in taxonomy and applied technologies, to allow for the mobilization and exchange of expertise, and to secure equitable resources.
- Investment in AI and automation must be coupled with equitable access and capacity-building.
- National initiatives (e.g., DiSSCo-UK) can serve as scalable models for regional and global collaboration.
- Long-term, multi-pillar programmes like SYNTHESYS offer a proven structure for enabling collaborative transformation in the sector.

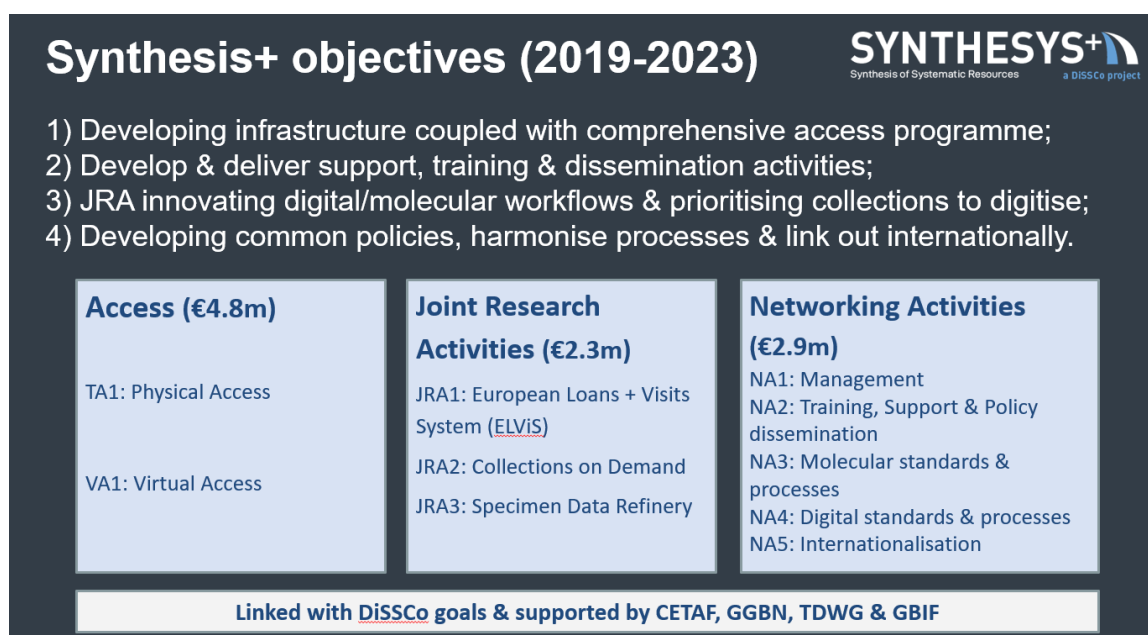


Figure 1: The Synthesis Model showcased as an example of a G20 collaborative model for NHCs, containing the building blocks of what collaborative actions for the Roadmap could look like

Some Comments from the Session

- There is a lot of data sitting with scientists and ecologists, and it's useful that these technological approaches like AI enable access to this. There is also a huge resource in the [Biodiversity Heritage Library](#), which is essentially the aggregator for literature around the natural world. An opportunity exists to mine this literature further.
- There is an opportunity for collaboration in dealing with **data quality, and especially with taxonomic names**, as many institutions only have pockets of taxonomic expertise. While initiatives like the Catalogue of Life exist to support high-quality access to species names, collaboration is key, so that countries can contribute to a global platform, and data quality related to taxonomic names can be addressed.
- With the volumes of data we are beginning to accumulate, AI also has the potential to transform the role of the taxonomist or data curators. Perhaps the future vision is not for taxonomists to curate the data, but rather to curate the models, rule sets, and algorithms that perform the cleaning for us. This approach maybe more sustainable in this data-driven world.
- GBIF (Christoph Hauser): Synthesis is one of the successes from Europe, in terms of bringing the community together. A question was posed regarding whether this model could be effectively adopted for capacity building in South Africa, and also in other regions of the world. Or if other mechanisms might be more efficient? If we were to look at enhancing/redesigning the Synthesis initiative, a value add would be to **support and bring together people who require capacity building related to technical skills and infrastructure, like those who run the technical digitization and collection labs**, or deal with aspects around taxonomy or taxidermy. In terms of efforts related to bilateral and political international cooperation, we could look at shared experiences and capacity building of technical skills. The focus should not be limited to research projects only, even though this is great. Outcomes should include publications, but also importantly shared experiences and the development of technical skills and capacity building.
- SANBI-GBIF: At national level, there are a number of elements in place to support a Synthesis type model (with the three tiers), supporting data and NHCs efforts. SANBI as an institution has a mandate to coordinate and manage biodiversity data and databases. Thus there is a strong policy approach and mandate. In terms of capacity building for the node we have a number of training initiatives. We have a holistic approach to capacity development, which also includes working with organizations, as well as universities, and developing post-graduate programs dealing with Biodiversity Informatics. The aim is for systemic impact. The NSCF also has a well-contained course dealing with NHCs, implemented with one of the universities. Network-related activities are also supported through SANBI-GBIF, the Foundational Biodiversity Information Programme, and the NSCF. While we have various efforts in place, we do need to scale up our activities at the national level, particularly in terms of infrastructure, to meet the needs of NHCs and the country's biodiversity data requirements. At the Africa level, the GBIF Nodes in Africa have been dealing with a regional approach in biodiversity informatics and have developed a business case for regional engagement. And what we have identified at this meeting is a need for further intra-African engagements. This includes the biodiversity informatics and NHCs community. Training should be systemic and also needs to happen at different levels, including technical and research levels.
- The networking activities have transactional costs, and there are costs to collaborate. **With Synthesis, there was a funding mechanism, so as we are thinking about a Roadmap, we must keep in mind that we need a funding mechanism to be in place to support this.**
- It was mentioned that the Natural History Museum of London publishes its business cases, as in the case of the Synthesis program, the grant proposal was published. A theme around business

case development also emerged, with the Africa Nodes also having drafted a business case for biodiversity informatics.

Table 1: Recommendations and Outcomes from the Seminar and Workshop on Global Collaborations Supporting Museums and NHCs

Thematic Area: Global Efforts Orchestration, Governance and Sustainability	Thematic Area: Regional Efforts Orchestration
R1: Establish Partnerships: • Purpose: To harness the power of NHCs and Biodiversity Informatics successful partnerships will need to be put in place. • Principles: Fund and create working groups across NHC globally. 1. Establish clear objectives and shared goals 2. Foster open communication and trust, 3. Utilise existing networks and platforms, 4. Develop roadmaps and action plans, 5. Focus on capacity building, and 6. Promote inclusivity and equity, 7. Sharing of best practice between the global north and south • Possible Partners: Museums, research organisations, Research-based networks (as CETAF), funders, industry, data science organisations, GBIF, government R2: Consider Long-Term Aspects In Sustaining Established and Well-Functioning Networks, Collect and Keep Contacts/Experiences From Past Networks • Purpose: learn about policy processes to find long-term solutions for infrastructures and networks • Principles: to have global infrastructures with regional and national nodes gives the infrastructures dimension that are difficult to shut down. Connect information about big global infrastructures • Possible Partners: NCBI, BOLD, GBIF, BIOBANKS for biology, GeoCASE for geosciences, connect with Mindat and other infrastructures R3: Strengthening GBIF: • Purpose: Strengthen and expand GBIF to all nations on a centralized platform for aligning national biodiversity data with global frameworks (CBD-GBF Targets, SDGs, etc.) and fostering international research collaborations. [This platform will also hold a list of experts dealing with measuring targets.] • Principles: Capacity development, Data standardization, open exchange, and strategic international partnerships, mobilisation of experts across countries	R7: Foster Regional Collaboration and Infrastructure To Develop Regional Hubs (Regional Natural History Infrastructure Network) • Purpose: To strengthen existing natural history institutions and integrate infrastructures developing technical and scientific hubs, to enhance regional biodiversity data collection, access and research. This will support species checklists, reference libraries and occurrences etc. • Principles: Collaborative development, regional focus, and leveraging diverse funding sources, leveraging existing tools, building upon shared digital infrastructures, and fostering collaborative organisations to address the current geographic fragmentation of collections and expertise. • Possible Partners: Natural History Museums (NHMs) across regions (e.g., SADC, East Africa, Asia), the private sector (for infrastructure investment), Research-based networks (as CETAF), and international research centres. Outcome: Developing regional hubs will deliver impact at a scale unachievable by isolated institutions, where Investment in the strategic organisation of collections and taxonomic capacity at a regional level is key. R8: Developing and Strengthening Regional Networks In Biodiversity Informatics • Purpose: Grow coordinated and empowered regional networks in biodiversity informatics. • Principles: Inclusivity, diversity, equity • Possible Partners: AU, EU, GBIF, GBIF Nodes, NHM London, CETAF, DiSSCO, Private Sectors, Funders, Government departments

• Possible Partners: Global Biodiversity Information Facility (GBIF), relevant UN agencies (e.g., CBD, SDGs), research institutions worldwide, and data science organizations. Outcome: A biodiversity data network which contributes to the achievement of GBF targets at national, regional and global level. R4: Creation Of International Working Groups On Specific Topics • Purpose: "To overcome knowledge gaps to leverage efforts" • Principles: symmetric baselines, international cooperation, • Possible Partners: Partnership between leading institutions on each specific topic, with related actors and interested stakeholders, including the use of existing working groups (like CETAF) for efforts like collections management, publishing, monitoring, and other domains. R5: Create Transnational Working Groups To Further Networking • Purpose: Create connecting nodes focused on themes within NHC or themes addressing biodiversity needs linked to NHC • Principles: Knowledge sharing; peer learning and support • Possible Partners: Curators, technicians, data managers, etc. from G20 countries' NHC R6. Sustainable Funding • Purpose: A Global Collections Collaboration initiative and associated activities requires investment to bridge capacity and infrastructure gaps between museums in the Global North and South. • Principles: equitable access between global north and south, inclusivity, transparency, show value, find (financial) support • Possible Partners: G20 support, Funders, governments, development banks, consultancy firms, industry, private sector engagement, bioscience companies, pharma companies.	R9: Utilize the Global Registry Of Scientific Collections (GRSciColl) To Develop An African and a Global Database Of Museum Natural History Collections • Purpose: To determine the number of natural history museums and collections on the African continent and assess what type of collections they house, the number of specimens housed by each collection and institution. This will help us understand what we have in an African context and mobilise resources and expertise to digitise data and develop the capacity of museum professionals through collaborative efforts on the continent • Principles: Transparency, professionalism, collaboration, honesty • Possible Partners: All Museums and other institutions with NHCs in Africa
Thematic Area: Indigenous Knowledge	Thematic Area: Transformation, Heritage and Culture (Looking back, acknowledging the past and moving forward)
R10: Indigenous Knowledge Systems & Ethics Initiative: • Purpose: To ethically integrate indigenous knowledge and traditional practices with scientific biodiversity data through community-led approaches.	R11: An Evolving Mandate for NHCs - Modernizing Collections & AI Integration • Purpose: transformation of NHCs institutions into dynamic hubs supporting the data, science-policy interface, that is also equipped to support policy and industry; To address museum backlog, safeguard

• Principles: Prior Informed Consent (PIC), reciprocal benefit, and cultural sensitivity. • Possible Partners: Indigenous communities, local traditional councils, legal and ethical review boards, and cultural heritage organizations.	existing collections, and leverage AI for advanced data analysis and global trend identification. • Principles: Socio-cultural transformation, Digital transformation, data interoperability, and responsible AI application. • Possible Partners: Museums, technology companies (AI/data solutions), funding bodies, national governments, and citizen science communities.
Thematic Area: Research	Thematic Area: Infrastructure
R12: Developing a Biodiversity Informatics Research Platform: • Purpose: To advance biodiversity informatics research and training by sharing expertise and resources. • Principles: Open access, mutual benefit, and systemic skill development. • Possible Partners: Research institutions, universities, national research foundations (e.g., NRF), and international biodiversity informatics initiatives. Outcome: Developing the field of biodiversity informatics R13: Develop New Or Modernised Research Infrastructure Like Joint Research Field Stations (Regional And International) • Purpose: Expanding knowledge of SA biodiversity, including descriptions of new taxa. Cross-transfer of knowledge and skills in biodiversity research • Principles: Application of genomic tools in addition to classical methods; training of a new cohort of taxonomists • Possible Partners: Russian and SA interested universities, NHCs, SANBI, GBIF	R14: Bottom-up infrastructure development • Purpose: Benefit from the regional and large national level funding and policy frameworks to self-organise at regional level. This can combine the benefits of a bottom up and top down approach. • Principles: Ensure interoperability with global infrastructures. E.g. GBIF, BOLD; Ensure use of common standards between regional infrastructures and networks; Create a federation approach linked to GBIF (think back on reviving Alliance for biodiversity); Look to existing regional initiatives for inspiration. • Possible Partners: CETAF, DiSSCo, GBIF, BOLD, large national infrastructures and other regional ones R15: Utilize the Global Registry Of Scientific Collections (GRSciColl) To Make Undigitized Collections Discoverable • Purpose: Get metadata of holdings of natural history institutions. This enables efficient collaborations, return of efforts in data analysis, and scientific accreditation • Principles: Use the global register of collections, where metadata is collected about the collections, the institutions holding them, contacts etc. • Possible Partners: GBIF.org, CETAF (with the Registry of Collections and GeoCASE for geological collections)
Thematic Area: Biobanks	
R16: Biobanking and Genetic Resources Collaboration • Purpose: To establish and expand biobanking facilities for genetic resources, supporting pharmaceutical, biotech, and other start-up industries. Biobanks can be expanded to include commercial focus, including developing a workflow.	

• Principles: Sustainable resource utilization, industry-academia collaboration, and ethical bioprospecting. Benefit sharing, job creation, ownership of IP • Possible Partners: Biotechnology companies, pharmaceutical industry, research universities, and national biobanks, Natural History Collections with tissue banks, culture collections.	
Thematic Area: Mobilising Data to fill data and knowledge Gaps	Thematic Area: Capacity Development
R17: New Technologies For The Enhanced Discovery Of Species, and the Crucial Role of Species Checklists and Reference Libraries • Purpose: The advent of new technologies, particularly high-throughput methodologies, offers transformative potential and should be strategically implemented. Image recognition, bioacoustic monitoring, and DNA-based evidence (e.g., eDNA metabarcoding) can generate vast biodiversity data. However, the utility of this data is fundamentally dependent on the availability and the robustness of reference libraries – curated databases that can accurately link these fragments of biodiversity evidence (an image, a sound, a DNA sequence) to a verified species name, robust backbones and nomenclatural registries. The creation and enrichment of these libraries are at the heart of biodiversity science. • Principles: Open access to publications, Data generation, reference libraries/checklists, high-throughput methodology • Possible Partners: Museums, Universities, technology companies (AI/data solutions), funding bodies, and citizen science platforms, NHM, Naturalis, DiSSCo, CETAF, GBIF, Funding Bodies - NSF, Darwin Initiative, GEF R18: AI-Driven Digitisation For Equitable Data Sharing: A Collaborative Initiative Leveraging Automated and AI-Powered Machine Learning Methodologies To Digitise and Disseminate High-Quality Data, Aiming To Bridge The Data Representation Gap Between The Global South and North • Purpose: To pilot scalable, AI-enhanced digitisation processes that ensure comprehensive, locally relevant data is accessible and shareable, thereby reducing disparities in global data representation and fostering inclusive knowledge exchange, supporting global science-policy outcomes for biodiversity conservation. • Principles: The project is grounded in principles of equity, transparency, collaboration, and cost-effectiveness, emphasising open access, ethical data	R21: Integrated Capacity Building Program: • Purpose: To build human (increase in number of personnel) and technical capacity (increase in skills) in biodiversity through structured training, internships, and permanent employment initiatives. Here capacity building will raise the next generation of NH researchers/curators, and enhance data sharing between institutions, to address emerging societal issues • Principles: Holistic development, modernising training including modern technologies, alignment with identified gaps, and governmental support for sustainability. • Possible Partners: Universities (for MSc/PhD programs), GBIF Nodes, vocational training centers, existing capacity-building initiatives (e.g., TBI Africa), and government ministries, NHC with global collections partners. R22: Building Capacity: Training Indigenous Representatives in Biodiversity Science to Enhance Representation and Indigenous Knowledge Access • Purpose: This would be a training initiative aimed at equipping Indigenous representatives with foundational and advanced skills in biodiversity-related disciplines. The objective is to improve Indigenous representation in environmental decision-making and scientific discourse, while also creating ethical and reciprocal pathways for incorporating Indigenous Knowledge Systems (IKS) into biodiversity research and conservation. • Principles: To provide targeted training in biodiversity monitoring, taxonomy, ecological research, and data stewardship to selected Indigenous participants. a.) To foster collaboration between Indigenous communities and scientific institutions in ways that honour Indigenous intellectual sovereignty. b.) To develop context-appropriate protocols for integrating IKS into biodiversity records and conservation strategies. c.) To

practices, capacity building, and shared financial responsibility to empower all participating nations. • Possible Partners: Members of regional networks from the global South and North who have experience, capacity and potential to undertake pilot projects. Outcomes: Mobilising foundational data from NHCs to fill the data and knowledge gaps, to support research, science and policy outcomes, realising the targets of the GBF. R19: Specimens and Data Should Be Together • Purpose: It is crucial that data and metadata of specimens, and the annotations of specimens are kept together, maintaining provenance information for all scientific information deriving from the specimens. Specimens are moved to the Internet as actionable and interoperable Digital Objects. • Principles: Embrace global persistent identification of the digital extended specimens and allocate a DOI for every specimens in the world. (Infrastructure already in place by DiSSCo available to all collections globally). Ensure citations are attributed to collections, Track citations across all uses of these DOIs and DOIs are issued free of charge to every collection • Possible Partners: DiSSCo - provides the open DOI infrastructure and data model for the digital specimens; GBIF - provides required global resources including GRSciColl; CETAF and other regional NH networks; TDWG - reflects developments in community standards; All individual collections R20: Linking All Specimen Data, Archives, Ecological Data For Coelacanths Between All South African Institutions Initially, But Expanding The System Later To Include All Specimens and Data Worldwide • Purpose: We have a lot of diverse samples and data and linking them using AI would be helpful for future access to information. It would make these data more available through a wider organisation group. If it works then the wish would be to expand it to other institutions/museums holding coelacanth specimens and data. • Principles: Better access to data using AI. • Possible Partners: SAIAB (specimens, data, archives), East London Museum (specimens and data), BMNH (AI skills).	promote long-term inclusion of Indigenous voices in biodiversity policy, research, and governance. • Possible Partners: Local (South African) Universities that can support both graduate and post-graduate training, and local institutions such as GBIF, NSCF, and government departments such as DSTI. Regionally, through CETAF- Distributed European School of Taxonomy (DEST) Academy and other linked resources hubs/platforms. Internationally, work with GBIF and their IKS project.
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Appendix 1: List of Participants

No.	Title	Name	Surname	Institute	Country
1	Ms	Noora	Albalooshi	Natural History Museum Abu Dhabi	UAE
2	Dr	Khadeeja	Alfaqeeh	Advanced Technology Research Council	UAE
3	Ms	Norah Abdulaziz	Alkhamis	Saudi Heritage Commission	The Kingdom of Saudi Arabia
4	Dr	Mohamed	Alkhemairi	Advanced Technology Research Council	UAE
5	Ms	Maitha	Almheiri	Ministry of Climate Change and Environment	UAE
6	Mr	Rashed	Alzaabi	Natural History Museum Abu Dhabi	UAE
7	Dr	Annemari	Antonites	Ditsong Museums of South Africa	South Africa
8	Dr	Minoli	Appalasamy	South African National Biodiversity Institute (SANBI)	South Africa
9	Prof	Alex	Asase	University of Ghana	Ghana
10	Mr	Ian Roger	Bills	South African Institute of Aquatic Biodiversity	South Africa
11	Dr	Wendy	Black	Iziko Museums of South Africa	South Africa
12	Prof	Hugo	de Boer	University of Oslo	Norway
13	Dr	Wayne Kevin	Florence	Iziko Museums of South Africa	South Africa
14	Dr	Theressa	Frantz	South African National Biodiversity Institute (SANBI)	South Africa
15	Ms	Anastasia	Gorbatko	Ministry of Science and Higher Education of the Russian Federation	Russia
16	Dr.	Christoph	Haeuser	Global Biodiversity Information Facility (GBIF)	Germany
17	Dr	Arindam	Bhattacharyya	Department of Science and Technology	India
18	Dr	Adriaana	Jacobs-Venter	Agricultural Research Council	South Africa
19	Mr	Adriaan	Jordaan	Natural Science Collections Facility/SANBI	South Africa
20	Ms	Jia Jun	Chua	Innovation, Research & Development Division, MTI	Singapore
21	Dr	Dimitrios	Koureas	Naturalis Biodiversity Center	Netherlands
22	Ms	Irina	Kuklina	International Centre for Innovation in Science, Technology and Education	Russia
23	Ms	Kim Elizabeth	Labuschagne	South African National Biodiversity Institute (SANBI)	South Africa
24	Mr	Mark	Lisher	Iziko Museums of South Africa	South Africa
25	Dr	Tim	Littlewood	Natural History Museum, London UK	UK

26	Dr	Anthony	Magee	Natural Science Collections Facility, South African National Biodiversity Institute	South Africa
27	Dr	Mudzuli	Mavhunga	SANBI - Biodiversity Biobanks South Africa	South Africa
28	Mr	Sibusiso	Mkhulisi	South African National Biodiversity Institute (SANBI)	South Africa
29	Dr	Joe	Miller	Global Biodiversity Information Facility	Norway
30	Mr	Shonisani	Munzhedzi	South African National Biodiversity Institute (SANBI)	South Africa
31	Dr	Geoffrey	Mwachala	National Museums of Kenya	Kenya
32	Ms	Martine Schulerud	Nilsen	Norwegian Ministry of Education and Research	Norway
33	Mr	Khanya	Ntondini	South African National Biodiversity Institute (SANBI)	South Africa
34	Ms	Ilze	Olckens	Tamarind Tree	South Africa
35	Dr	Fatima	Parker-Allie	South African National Biodiversity Institute (SANBI)	South Africa
36	Ms	Desiree	Paulsen	Tamarind Tree	South Africa
37	Mr	Daan	Du Toit	Department of Science, Technology and Innovation	South Africa
38	Ms	Shi'an	Tay	Innovation, Research & Development Division, MTI	Singapore
39	Mr	Dmitry	Shkolyar	Embassy of the Russian Federation	Russia
40	Dr	Vince	Smith	Natural History Museum, UK	UK
41	Dr	Pia Angela	Stieger Cecchini	Swiss Academy of Sciences	Switzerland
42	Mr	Andrei	Tolstikov	University of Tyumen	Russia
43	Ms	Saori	Utsunomiya	Secretariat of Science, Technology and Innovation Policy, Cabinet Office, Japan	Japan
44	Mr	Edwin	Van Huis	Naturalis Biodiversity Center	Netherlands
45	Mr	Erich	Van Wyk	South African National Biodiversity Institute (SANBI)	South Africa
46	Mr	Kazuhisa	Yoshida	Secretariat of Science, Technology and Innovation Policy, Cabinet Office, Japan	Japan
47	Dr	Wei	Zhang	Shanghai Science & Technology Museum	China
48	Mrs	Ana	Casino	CETAF	Belgium
49	Dr	Vince	Smith	Natural History Museum, UK	UK
50	Mr	Jank	Bujwen	Ministry of Science ICT, Korea	South Korea

51	Mr	Park	Hwanzl	STEPI	South Korea
52	Ms	Kwon	Sohyun	STEPI	South Korea
53	Prof	Daniel	Parker	University of Mpumalanga	South Africa
54	Prof	Aaron	Tshizumba	University of South Africa	South Africa
55	Mr	Selby	Modiba	Department of Science, Technology and Innovation	South Africa
56	Ms	Mmampei	Chaba	Department of Science, Technology and Innovation	South Africa
57	Mr	Leluma	Matooane	Department of Science, Technology and Innovation	South Africa
58	Ms	Lindiwe	Gama	Department of Science, Technology and Innovation	South Africa
59	Mr	Kagiso	Moloko	Department of Science, Technology and Innovation	South Africa

Appendix 2: G20 Research and Innovation Working Group (RIWG) Seminar Agenda



Developing Global Collaborations to Support Museums and Natural History Collections

28th May 2025 – Multipurpose Hall, University of Mpumalanga, Mbombela

Time: 9am -1pm

Half-Day Seminar with 2 sessions, 90 minutes each

Welcome and Opening Remarks:

- Mr. Shonisani Munzhedzi, CEO, South African National Biodiversity Institute
- Joe Miller – GBIF Executive Secretary

Session 1: Global Collections: The Imperative for Action Now

Purpose: This session sets the scene for the current state of play in natural history collections (NHCs). We will look at existing activities, priorities and challenges and the value of collections to science, society and the bioeconomy. The session will identify some of the asymmetries as outlined in the deliverables (and the background document) and will identify potential opportunities to forge new partnerships.

Speakers and panellists will come from South Africa, G20 countries and other African nations

Format: 3 Short keynotes, (15 minutes each)

- Tim Littlewood - The Global Imperative - the role of NHCs in providing solutions to global challenges
- Anthony Magee – Natural Science Collections Facility: The current state of collections in South Africa
- Álex Asase - NHCs perspectives from Africa

Panel Discussion (30 mins): The Imperative For Action From The Perspective Of Science Policy & Funders

Panel: Tim Littlewood, Anthony Magee, Alex Asase, Edwin van Huis, DSTI

Possible questions posed from the facilitator or the audience:

- How can global collaboration advance science, technology and innovation through key research infrastructures, like natural history collections?
- How can the socio-economic value of Natural History Collections be better articulated to policymakers and the public?
- Mobilising priority biodiversity datasets are critical for filling data and knowledge gaps, to support key research questions and for science-based decision-making. What are the potential funding opportunities to support the science-policy outcomes and to mitigate the current planetary crisis?

Tea/Coffee Break

Session 2: Global Collections: Securing Our Future, Starting Today

Purpose: Natural history collections, as repositories of invaluable biological information and as distributed scientific infrastructure with the capacity to actively engage with and support solutions-driven science, are uniquely placed to address urgent and emergent issues through research, education, and conservation. We will cover topics like Indigenous Knowledge Systems and their applications in NHCs future thinking. We aim to identify collaborative actions that can help support global collaboration in the Natural History Collections arena.

Speakers and panellists will come from South Africa, G20 countries and other African nations

Format: 4 Short Keynotes (15 minutes each)

- Talks:
 - Christoph Hauser - Natural History Collections for the 21st Century: Scalable, Open, and AI-Ready Workflows for Biodiversity Discovery
 - Wayne Florence- Iziko Museums of South Africa at 200: Integrating Science, Culture, and Indigenous Knowledge for a More Equitable and Innovative Future
 - Geoffrey Mwachala – Perspectives from Africa
 - Fatima Parker-Allie - From Innovation to Impact: Advancing Biodiversity Informatics for a Sustainable Earth

Panel: Christoph Hauser, Wayne Florence, Vince Smith, Geoffrey Mwachala, and Fatima Parker-Allie

Panel Discussion (30 mins): Meeting 21st Century Challenges

Possible Leading questions posed from the facilitator and by the audience:

- How can collaboration rebalance the current asymmetries?
- What are the ethical considerations when incorporating IKS into global biodiversity data infrastructures?
- How can collections advance innovative technological infrastructure development while growing the human capacity?
- What are the barriers to adopting advanced digitisation and bioinformatics tools in developing nations, and how can they be overcome?

The Seminar will be followed by a safari and braai in the afternoon

Output/Deliverable

Summary report/roadmap on the future of collaborations will be developed with input from all speakers and audience.

Facilitators

Organisational transformation facilitators will moderate the panels.

Audience

The Seminar will take place as part of the programme under the RIWG meeting. It will include the RIWG and museum experts nominated by the countries to attend.

Appendix 3: G20 Research and Innovation Working Group Workshop Agenda



Developing Global Collaborations to Support Museums and Natural History Collections 29 May 2025, Multi-Purpose Hall, University of Mpumalanga, Mbombela, Mpumalanga Province

8.30 - 8.40	Welcome and Opening of the Workshop	SANBI
8.40 - 8.55	Uncovering our Common Intent Introducing the U-process and Workshop Technologies to support our outcomes	Facilitators
8.55 – 9.30	Bridging Asymmetries: Building Equitable and Sustainable Partnerships for the Future of Natural History Collections Provocation 5 min Conversation in Pairs 10 min / Pairs partnered 10 min	Facilitators
9.30-10.30	Sensing 1: New Roles for Old Institutions: Transforming the Museums and Natural History Collections Landscape Bending the curve through collaboration with policy and industry Keynote with facilitated Responses	Facilitators & Speakers Edwin van Huis (Naturalis Biodiversity Center, CETAF)
10.30 – 11.00	Refreshments	
11.00 -11.50	Sensing 2: Developing Initiatives and Building Collaborations at National, Regional and Global Levels with a focus on data mobilisation - Building a continental-scale infrastructure in support of collections-to-impact value chains. - A bottom-up initiative mobilising knowledge from natural history collections in Switzerland - The SANCF and Microbial Collections across Africa-working towards the "African MCCs Consortium". Presenters 10 min max Input followed by TRIAD conversations with Focus Questions 20 min	Facilitators & Speakers Dimitris Koureas (Naturalis Biodiversity Center; DISSCO) Pia Stieger (Swiss Academy of Sciences, SwissCollNet) Adriana Jacobs Venter (Agricultural Research Council)

11h50 – 12h30	Sensing 3: Building Capacity in the Natural History Collections Community to Support Biodiversity Research and Transdisciplinary Collaboration • Staff training and taxonomic impediment – how collaborations can help. • Taking the Collections Out of the Museum: Supporting Biodiversity Research and Transdisciplinary Collaboration. Flash Talks followed by TRIAD conversation with Focus Questions 20 min Presenters 5 min max	Facilitators & Speakers Roger Bills (SAIAB) Hugo de Boer (University of Oslo)
12h30 – 13h00	Sensing 4: Exploring Mechanisms Among Global Natural History Collections for Inclusive Community Consultation and Multidisciplinary Data (Including indigenous knowledge) Integration • Without ethics, inclusion is not integration. • GBIF's efforts with Indigenous Knowledge Flash Talks followed by Pair Conversation with Focus Questions 10 min Presenters 5 min max	Facilitators & Speakers Wendy Black (Iziko Museums) Joe Miller (GBIF)
13.00 – 14.00	Lunch	
14.00- 14.45	Integrating Infrastructure and E-Infrastructure: Advancing Natural History Collections into the Digital Age Session Lead: Vince Smith (UK Natural History Museum) with Joe Miller (GBIF)	Vince Smith
14.45 – 15.15	Facilitated Open Space	Facilitators
15.15– 16.00	Working Refreshments Crystallising Equitable & Sustainable Collaborative Opportunities – Facilitated process	Facilitators
16.00 – 16.30	First Iteration of Prototyping of Collaborative Opportunities: Towards a Roadmap – Facilitated Process	Facilitators
16.30 – 17.00	Closing Circle	Facilitators GBIF/SANBI

Photo Gallery

